Human-Centric Adaptive Assessment: Evaluating Static Rule-Based and Dynamic LLM-Based Mental Health Questionnaires

Jaeyun Ree, Khang Huynh, and Zachary Yeap
Department of Software Engineering

Abstract—**Static mental health check-in questionnaires can be clear and clinically grounded, but they often repeat the same questions regardless of a user's fluctuating mental health state. This study presents Navigator, a web-based mental health questionnaire prototype comparing a static rule-based monthly check-in with a dynamic LLM-based method. This dynamic method adapts the baseline questions using a controlled persona and a health profile from the previous month. Forty valid participants evaluated both methods through a within-subjects survey grounded in COSMIN content-validity dimensions: Clarity, Relevance, and Quality. The LLM-based method received higher composite ratings across all three dimensions, with the most pronounced difference in Relevance. Qualitative responses further suggested that participants valued the LLM method for its personalisation and history-aware questioning, while the rule-based method remained understandable and straightforward. These findings indicate that the primary advantage of LLM-based adaptation lies not in clearer wording alone, but in making questions feel more contextually relevant to an evolving health profile, and that leveraging longitudinal assessment data to improve content validity may support more accurate health outcome prediction.**

I. Introduction

Mental health is increasingly recognised as a critical component of overall well-being, and digital health applications are playing a growing role in routine screening and self-monitoring, such as periodic mood check-ins and longitudinal symptom tracking. In practice, most of these applications rely on standardised static instruments such as the PHQ-9 [1] and the GAD-7 [2], which present the same fixed set of questions at every check-in regardless of the user's current condition, prior responses, or longitudinal history. While these instruments are clinically validated, repeated administration of the same questions can increase self-report burden and reduce user engagement over time [3].

Adaptive questionnaire techniques, such as pre-determined branching logic and computerised adaptive screening, have been introduced to partially address this limitation by tailoring question sequences to user input [4]. However, these static rule-based mechanisms remain bounded by predefined decision trees and cannot generate non-repetitive new question sets or dynamically reframe questions based on a user's individual context. Prior work on computerised adaptive testing has shown that adaptive selection can reduce question burden while preserving the quality of the measurement [5], [6], further motivating exploration of more flexible adaptation methods [7].

Recent advances in Large Language Models (LLMs) open a new design space for adaptive assessment: LLMs can synthesise context-aware questions informed by a user's health profile, lifestyle changes, and prior responses, and have begun to be explored as engines for patient-reported outcome measures [8]. Despite this promise, the use of LLMs in mental health remains at an early stage. A recent systematic review of mental health chatbots highlights that most existing systems use pre-determined rule-based questionnaires, and that LLM-based questionnaire designs, while more flexible, introduce new concerns around safety, appropriateness, and clinical reliability [9], [10], [11]. Despite this growing body of work, no study has directly compared LLM-based and rule-based adaptive mental health questionnaires within a single application using a controlled, human-centric evaluation grounded in established content-validity criteria such as clinical reliability.

To address this gap, we developed *Navigator*, a web-based prototype that implements two distinct monthly mental health check-in methods within the same application environment:

- **Static rule-based method:** predefined branching logic determines which follow-up questions appear based on fixed risk-level thresholds.
- **Dynamic LLM-based method:** Google Gemini 2.5 Flash adapts baseline questions using the user's health profile and prior assessment data.

Using a within-subjects study design, participants interacted with both methods via a standardised persona (Sophie) and completed a structured feedback survey. Evaluation metrics are grounded in the COSMIN content-validity framework [12], which identifies *Clarity*, *Quality*, and *Relevance* as core dimensions of questionnaire content validity.

This study aims to answer the following research questions:

- **RQ1 (Clarity):** Do LLM-based questionnaires produce higher perceived clarity than rule-based questionnaires?
- **RQ2 (Quality):** Do LLM-based questionnaires produce higher perceived quality than rule-based questionnaires?
- **RQ3 (Relevance):** Do LLM-based questionnaires produce higher perceived relevance than rule-based questionnaires?

The main contributions of this work are: (1) the Navigator prototype, a Flutter web application implementing both rule-based and LLM-based adaptive questionnaires within a unified interface; (2) a within-subjects user study design using a standardised persona to ensure fair comparison across conditions; and (3) an empirical, COSMIN-grounded evaluation of user perceptions of Clarity, Quality, and Relevance for LLM-based versus rule-based adaptive mental health questionnaires.

The remainder of this paper is organised as follows. Section II reviews related work on mental health questionnaire systems, LLMs in healthcare, and adaptive survey design. Section III presents the methodology, covering the research approach, system design, controlled scenario, study design, and evaluation metrics. Section IV presents the results, followed by discussion in Section V, and conclusions in Section VI.

II. Related Work

Digital mental health applications increasingly offer routine self-monitoring through periodic check-ins, yet most rely on static, validated instruments that repeat the same items regardless of a user's changing condition [1], [2]. Adaptive approaches, from computerised adaptive testing to rule-based branching interfaces, have partially addressed this rigidity, but remain bounded by predefined item banks or decision trees [5], [4]. More recently, large language models have been explored as engines for generating personalised patient-reported outcome items [8], though their use in mental health is still at an early stage and raises concerns around safety and clinical reliability [9], [11]. Despite this growing body of work, no study has directly compared rule-based and LLM-based adaptive questionnaires within a single application using a controlled, human-centric evaluation.

Navigator is a web-based mental health questionnaire prototype developed to address this gap. It implements both a static rule-based and a dynamic LLM-based monthly check-in method within the same application environment, allowing direct comparison under identical conditions. Although the broader project was initially framed around adaptive assessment across multiple chronic-disease domains [13], this paper narrows the evaluation to mental health check-ins, which provide a concrete, high-impact, and survey-feasible context.

We accordingly organise prior work along three lines that progressively narrow toward our research question. We first review the dominant paradigm of static, validated mental health instruments and the limitations they impose on longitudinal self-report. We then survey existing adaptive questionnaire strategies, from computerised adaptive testing to rule-based interface adaptation, and most recently to LLM-based patient-reported outcome measures. Finally, we examine the emerging body of work on LLM-based mental health systems, identifying both their promise and the safety considerations that constrain their design.

A. Mental Health Assessment and Static Instruments

Mental health screening in primary care and digital health applications has long relied on standardised, fixed-item self-report instruments. The Patient Health Questionnaire-9 (PHQ-9) for depression [1] and the Generalised Anxiety Disorder-7 (GAD-7) scale [2] remain among the most widely used tools, demonstrating strong sensitivity and specificity (both 88% for PHQ-9 $\geq$ 10; 89% and 82% for GAD-7 $\geq$ 10) and providing interpretable severity cut-offs across diverse populations. Their brevity, transparency, and validated scoring make them attractive for integration into digital platforms aimed at routine, longitudinal mental health monitoring.

However, the very stability that gives these instruments their psychometric strength also imposes meaningful limitations on user experience. Static questionnaires repeat the same items at every administration, regardless of a user's current state or response history. Longitudinal self-report can suffer from fatigue and completion challenges when users complete the same scale weekly or monthly. Computerised adaptive testing studies have shown that adaptive item selection can reduce item burden while preserving depressive-symptom measurement performance [5], [6], and recent work has argued for more adaptive health questionnaires [7]. Ecological momentary assessment studies have documented this problem directly: in a 28-day smartphone-based depression study, only seven of eleven participants completed the full protocol and the average daily response rate was 62.5% [3], suggesting that repeated digital self-report can suffer from engagement and completion challenges, even when the underlying measures are clinically meaningful.

Beyond engagement, fixed wording can raise concerns about content validity and relevance. Baas et al. [14] demonstrated that one PHQ-9 item exhibited measurement non-invariance across ethnic groups, revealing cultural bias in supposedly universal items. Wild et al. [15] similarly argue that questionnaire wording must be treated as a structured design and validation problem rather than a literal translation. These findings indicate that even well-validated instruments carry implicit assumptions about who the respondent is and how they interpret each item. Adaptive systems may help identify and mitigate these assumptions through more context-sensitive wording and item selection.

B. Adaptive Questionnaire Approaches

To mitigate the burden and rigidity of static instruments, prior work has explored adaptive questioning strategies. The most established approach is Computerised Adaptive Testing (CAT), which uses item response theory to select the most informative items based on prior responses. Gibbons et al. [5] demonstrated that the CAT-based assessment of mood and anxiety disorders reduced the number of administered items by approximately 95% relative to a full item bank, while preserving strong correlation with the full-scale score. Choi et al. [6] likewise compared static short forms, two-stage branching, and CAT against the full PROMIS depression item bank, finding that CAT outperformed static short forms on most performance criteria, with two-stage semi-adaptive testing approaching CAT-level performance.

A parallel line of work has examined rule-based adaptive interfaces in broader health contexts. In a systematic review of adaptive user interfaces for chronic disease, Wang et al. [4] identify three dominant adaptation strategies: rule-based "if–then" logic, machine-learning-driven prediction, and feedback-loop systems that revise content based on accumulating user data. Although rule-based approaches dominate the surveyed literature due to their interpretability and low computational overhead, they are inherently bounded by the comprehensiveness of their predefined rules and cannot adapt to contexts the designer did not anticipate.

A more recent direction extends adaptation beyond item selection toward item generation. Terheyden et al. [8] introduce the concept of LLM-powered Patient-Reported Outcome Measures (LLM-PROMs), in which large language models generate personalised open-ended items and interpret responses against established psychometric scaffolds. Their framework argues that LLM-based PROMs should be developed from validated content and benchmarked against fixed instruments to preserve content validity. This conceptual shift, from *selecting* items in a finite bank to *generating* items conditioned on user context, defines the design space we examine in this paper. Unlike CAT, which adapts by selecting from a validated item bank, LLM-based methods can also alter wording or generate new follow-up prompts, creating both opportunities for personalisation and additional risks for validity.

C. LLM-Based Systems in Mental Health

The emergence of large language models has accelerated interest in conversational and adaptive mental health tools, but evidence on their efficacy and safety remains preliminary. In the most comprehensive review to date, Hua et al. [9] surveyed 160 mental health chatbot studies published between 2020 and 2024, classifying them as rule-based, machine-learning-based, or LLM-based. Their analysis reveals an important asymmetry: although LLM-based systems grew rapidly in 2024, they remain concentrated in early-stage validation, while rule-based systems still dominate later-stage human and clinical testing. This positioning motivates our study as a pilot, user-centred feasibility evaluation rather than a clinical efficacy trial.

User-facing evidence further suggests that LLM behaviour in mental health is highly condition-dependent. Li et al. [10] analysed a large dataset of online discussions about LLM use for mental health and reported condition-specific differences in user sentiment and concerns. This indicates that LLM acceptability should not be treated as uniform across mental health contexts. Complementing this, Shen et al. [11] conducted a clinician-blind evaluation of multiple LLM versions on 158 prompts and found that no tested model reliably produced appropriate responses to psychotic content. Together, these findings establish two requirements for any LLM-based mental health tool: condition-aware framing and explicit safety boundaries. This distinction is also important for our setting: questionnaire systems do not aim to provide therapeutic dialogue, but rather to elicit structured, interpretable self-reports through clear and relevant prompts. The risks identified in conversational LLM use therefore translate, but do not transfer directly, into the design constraints for LLM-based questionnaire items.

Despite this growing body of work, limited prior research has directly compared rule-based and LLM-based adaptive questionnaires within a unified application using a controlled, within-subjects evaluation. Existing comparisons either contrast different chatbot generations against one another [9], study LLM use in unstructured user discussions [10], or evaluate adaptive item *selection* without addressing LLM-based *wording* [5], [6]. Our work addresses this gap by holding the application environment, the evaluation persona, and the participant group constant, thereby isolating the effect of the adaptation method itself.

Although several evaluation approaches exist for patient-reported outcome measures, the COSMIN (COnsensus-based Standards for the selection of health Measurement INstruments) framework [12] is widely recognised as the methodological standard for assessing content validity. Drawing on an international Delphi study of 159 experts, COSMIN formalises content validity along three dimensions: *relevance*, *comprehensiveness*, and *comprehensibility*. These dimensions align directly with the evaluation needs of this study, as described in Section III-H.

III. Methodology

This section presents the end-to-end methodology used in this study. The research proceeded through six stages: (1) a literature review of static and adaptive mental health instruments to identify the limitations that motivate this work (Section II); (2) identification of validated questionnaire items, drawing primarily on the PHQ-9 [1] and GAD-7 [2], and a decision to focus the evaluation on the mental health domain; (3) design and implementation of two adaptation methods, rule-based and LLM-based, within a unified web-based prototype (Sections III-A–III-C); (4) construction of a controlled evaluation scenario using a standardised persona (Section III-D); (5) design of a within-subjects user study with a COSMIN-grounded feedback survey (Sections III-E–III-H); and (6) data collection and analysis (Section III-I).

The prototype was designed as an evaluation instrument rather than a production system: its purpose is to present both methods within a single, controlled application environment so that user perceptions can be compared fairly. Architectural choices, including a shared interface, fixed persona, and identical health domains, were therefore driven by the evaluation design.

A. Architecture Overview

Navigator is implemented as a Flutter web application backed by Supabase (PostgreSQL) for question storage and branching-rule definitions. A single-page interface presents foundational intake questions, monthly check-ins for six health domains (Mental Health, Dietary, Physical Activity, Alcohol, Smoking/Vaping, and Women's Health), and a results dashboard. Although the broader prototype supports all six domains, the present study evaluates only the Mental Health monthly check-in pathway, which provides a concrete and high-impact context for comparing adaptation methods. Both the rule-based and LLM-based monthly check-in methods share the same front-end layout, question rendering components, and scoring pipeline, so that any difference participants observe can be attributed to the adaptation method itself rather than to interface variation.

For LLM-based question generation, the application calls the Google Gemini 2.5 Flash API. This model was selected for its low latency and structured-output support (native JSON response mode), which are practical requirements for generating questionnaire items in real time within a web application. A deterministic fallback path returns pre-authored enhanced questions if the API is unavailable, ensuring that every participant in the study receives the same enhanced questionnaire content regardless of network conditions.

B. Static Rule-Based Adaption Method

The rule-based method adapts monthly check-in questionnaires through a branching engine that evaluates predefined conditional rules after each user response. Each rule is a triple of *trigger question*, *boolean expression*, and *target action*. When a user answers the trigger question, the engine evaluates the expression against the response value; if the condition is met, the target questions are shown or hidden accordingly. This approach mirrors the "if–then" rule-based adaptation strategy identified as the dominant paradigm in chronic-disease adaptive interfaces [4].

Rules operate on single-question, single-timepoint data: a rule can reference the current answer to one question but cannot access prior month responses or aggregate longitudinal patterns. For the Mental Health domain, risk levels are classified at the domain level using project-specific percentage-based thresholds loosely inspired by the PHQ-9 severity bands [1]: scores below 25% of the domain maximum are classified as *None*, 25–50% as *Mild*, 50–75% as *Moderate*, and above 75% as *High*. These thresholds are a simplified mapping for the prototype and do not correspond directly to the validated PHQ-9 cut-off scores.

A concrete example illustrates the method's limitation. Table I lists the mental health questions presented under each method in Sophie's scenario (Section III-D). Under the rule-based method, Rule 13 specifies that the functional-impact question is shown only when the mood frequency response is ≥ 2 on a 0–3 scale. Sophie's Month 2 response of 1 ("Several days") does not meet this threshold, so the functional-impact item remains hidden, even though two consecutive months of Mild risk may warrant exploring it. The rule-based system cannot detect this longitudinal pattern because each rule evaluates a single response in isolation.

C. Dynamic LLM-Based Adaption Method

The LLM-based method adapts personalised monthly check-in questions by conditioning a large language model on the user's prior health snapshot. At the start of a monthly check-in, the system constructs a structured prompt containing two inputs: (1) a *UserSeed*, a versioned summary of the user's demographics, domain-level risk classifications, and free-text context notes from the previous assessment period; and (2) the baseline rule-based questions for the current domain, so the model understands what it is improving upon.

The prompt instructs the model to produce questions that are *more relevant, clearer, and less repetitive* than the rule-based baseline, grounding the generation task in the same content-validity dimensions (Clarity, Quality, Relevance) used in the evaluation (Section III-H). The model returns a JSON array of question objects, each containing the question text, response options, and a brief rationale for the adaptation. Generation uses a low temperature setting ($T = 0.3$) to favour consistency and structured output over creative variation.

This design aligns with the LLM-PROM framework proposed by Terheyden et al. [8], which argues that LLM-generated patient-reported outcome items should be developed from validated content and benchmarked against fixed instruments. In Navigator, the LLM does not generate questions from scratch; it *transforms* baseline questions derived from validated instruments together with project-specific follow-up items, using the user's longitudinal context. Table I presents the five specific transformations applied in Sophie's scenario: history-aware rephrasing of the mood frequency item, occupation-specific language for the manageability item, a new trend-direction item that captures the two-month persistent Mild pattern, early unlocking of the functional-impact item based on longitudinal rather than single-month thresholds, and omission of the isolation item where Month 1 data showed no change was expected.

D. Controlled Scenario: Sophie

To enable a fair within-subjects comparison, both methods are evaluated using a single standardised persona rather than participants' own health data. "Sophie" is a fictional 24-year-old female East Asian international university student whose Month 1 foundational assessment produced a Mild mental health risk prediction and None across all other domains. Her context notes record specific response details: a mood frequency score of 1 ("Several days"), a low and stable isolation score, and no triggered functional-impact questions.

The decision to use a fixed persona serves three evaluation purposes. First, it eliminates between-participant variation in health profiles, ensuring that every participant sees the same rule-based and LLM-based questionnaires and that observed differences in Clarity, Quality, and Relevance ratings reflect the adaptation method rather than individual health circumstances. Second, it avoids the ethical complications of collecting real mental health data in a prototype evaluation context. Third, it provides a scenario that concretely illustrates the limitation of rule-based adaptation: Sophie's profile is designed so that the

TABLE I

MENTAL HEALTH CHECK-IN QUESTIONS UNDER EACH ADAPTATION METHOD FOR SOPHIE'S MONTH 2 SCENARIO, MAPPED TO COSMIN CONTENT-VALIDITY DIMENSIONS. THE RULE-BASED COLUMN SHOWS THE QUESTIONS PRODUCED BY THE BRANCHING ENGINE; THE LLM-BASED COLUMN SHOWS THE GEMINI 2.5 FLASH-ASSISTED TRANSFORMATIONS.

Item	**COSMIN**	**Rule-Based**	**LLM-Based**	**Transformation Rationale**
Mood frequency	Clarity	"In the past month, how often have you felt low mood, sad, hopeless, nervous, worried, or restless?"	"Over the past month, have you continued to experience feelings like low mood, worry, or restlessness? Last month you reported this happening on several days."	References Month 1 history; acknowledges continuity rather than repeating identical wording.
Managing life	Clarity	"…how often have you felt that important things in your life were difficult to manage?"	"…has it felt harder than usual to stay on top of your studies, daily responsibilities, or relationships?"	Replaces generic "important things" with student-specific language.
Trend direction	Relevance	Not asked	"Since you've reported similar feelings for two months in a row, how would you describe the change over the past month?"	New item capturing longitudinal signal (two-month Mild pattern) that rule-based thresholds cannot detect.
Functional impact	Quality	*Hidden* because the mood score of 1 does not meet the ≥ 2 threshold required by the branching rule.	"Have these feelings made it harder for you to manage your daily life, for example, attending classes, completing work, or maintaining relationships?"	Unlocked early based on persistent pattern; phrasing adapted to student context.
Isolation	Relevance	"In the past month, how often do you feel isolated from others?"	*Skipped* because the Month 1 score was low and stable, so no new information was expected.	De-prioritises items unlikely to yield new signal, reducing respondent burden.

branching rule's hard threshold blocks the functional-impact question despite a clinically meaningful two-month pattern (see Table I), creating a clear point of contrast for participants to evaluate.

The persona was designed with attention to measurement validity considerations identified in prior work. Baas et al. [14] demonstrated that PHQ-9 responses can exhibit measurement non-invariance across demographic groups, and Wild et al. [15] emphasised that questionnaire wording must account for cultural context. Sophie's profile, an East Asian international student, was therefore chosen to represent a demographic for whom culturally adapted wording (as produced by the LLM method) might be particularly relevant, while remaining relatable to the predominantly university-aged participant pool.

E. Study Design

This study uses a within-subjects design: every participant interacts with both the rule-based and the LLM-based monthly check-in methods and then rates each on the same set of evaluation items. The within-subjects approach was chosen because it allows each participant to serve as their own control, increasing statistical sensitivity to differences between methods while requiring fewer participants than a between-subjects design.

Participants completed the rule-based check-in first, followed by the LLM-based check-in. This fixed presentation order was adopted because the rule-based condition establishes a baseline context that participants can reference when evaluating the LLM-based adaptations; the LLM method is designed to improve upon the specific limitations of the rule-based output (see Table I), and viewing the baseline first makes these improvements interpretable. Counterbalancing was not implemented. We acknowledge this as a limitation: order effects cannot be fully ruled out, and this is discussed further in Section V-C.

The study was approved by the Monash University Human Research Ethics Committee (MUHREC) under a Non-HREC (Lower Risk) review (Project ID 51664). All participants were fully informed of the study's purpose and provided consent via a written explanatory statement before beginning.

F. Participants

Participants were recruited through convenience sampling via direct contact with peers, classmates, and personal networks. Eligible participants were adults aged 18 years or older, comprising university students and working professionals. No monetary compensation or reimbursement was offered. In accordance with the ethics approval, students enrolled at Monash University Malaysia and Indonesia campuses were excluded from recruitment, as the approval was scoped to the Australian campus.

A total of 42 responses were received between 25 April and 22 May 2026. After excluding one non-consent response and one invalid submission, 40 valid responses were retained. Demographic details are reported in Section IV-A.

G. Procedure

Each participant completed the study independently in a single session, following a four-step procedure:

1) **Orientation.** Participants received a link to the Navigator prototype along with a user guide explaining the application interface and the Sophie persona. The guide described Sophie's demographics, her Month 1 health snapshot, and the purpose of the monthly check-in. Participants were instructed to evaluate the questionnaire from Sophie's perspective.
2) **Rule-based check-in.** Participants completed Sophie's Month 2 mental health check-in using the rule-based method. The branching engine determined which follow-up questions appeared based on their responses to the pre-filled baseline items. Three questions were presented; the functional-impact item remained hidden because the branching rule's threshold was not met (Table I).
3) **LLM-based check-in.** Participants then completed the same Month 2 check-in using the LLM-based method, which presented transformed questions adapted using Sophie's longitudinal profile. The interface layout and interaction pattern were identical to the rule-based condition.
4) **Feedback survey.** After completing both check-ins, participants were directed to an external Google Form survey comprising five sections: *Section A* (Demographics) collected age range, gender, health-app usage frequency, and student/professional status; *Section B* (Clarity), *Section C* (Relevance), and *Section D* (Quality) each contained three 5-point Likert items, where every item was rated for both the rule-based and LLM-based methods side by side; *Section E* (Overall Experience) asked participants to endorse one or more satisfaction options (rule-based, LLM-based, both, or neither) and to provide free-text feedback. Sections B and D also included an open-ended question each, yielding three open-ended prompts in total.

The entire session, including prototype interaction and survey completion, was estimated to take 10–15 minutes.

H. Evaluation Metrics

We adopt the COSMIN content-validity framework [12] to evaluate the two adaptation methods on common ground. COSMIN was selected because it is the most widely adopted standard for assessing the content validity of patient-reported outcome measures, grounded in an international Delphi consensus of 159 experts. This provides a principled basis for comparison that draws on established methodology rather than ad-hoc usability heuristics. We map the three COSMIN dimensions onto study-specific evaluation metrics as follows:

- **Clarity** (COSMIN *comprehensibility*): whether the questionnaire wording is clear and easy to understand.
- **Quality** (COSMIN *comprehensiveness*): whether the questions are well-formed, meaningful, and complete within the monthly check-in context. Participants judge whether each adaptive question appears appropriate and useful.
- **Relevance** (COSMIN *relevance*): whether the questions are appropriate and pertinent to Sophie's health profile and circumstances.

Each dimension was measured using three Likert-scale items rated from 1 (*Strongly Disagree*) to 5 (*Strongly Agree*). Each item was presented as a side-by-side matrix in which participants rated both the rule-based and LLM-based methods on the same screen, yielding nine paired ratings (18 data points) per participant. The three items within each dimension were averaged to produce a single composite score for Clarity, Quality, and Relevance for each method.

In addition, three open-ended questions were embedded across the survey sections: one within the Clarity section (asking about confusing or unclear questions), one within the Quality section (asking participants to compare how each version felt to complete), and one in the Overall Experience section (inviting general feedback on both methods). These qualitative responses supplement the Likert data by surfacing themes that structured items may not capture. A final satisfaction item asked participants to endorse one or more of four options: rule-based method, LLM-based method, both, or neither.

I. Data Collection and Analysis

Survey responses were collected via Google Forms between 25 April and 22 May 2026, yielding 42 submissions. Two responses were excluded during data cleaning: one participant did not provide consent, and one submission was identified as invalid based on uniform extreme ratings across all 18 Likert items combined with non-substantive open-ended answers. The final analysis sample comprised $n = 40$ valid responses.

For quantitative analysis, paired-sample t-tests were used to compare composite scores between the two methods for each COSMIN dimension. Effect sizes were computed using Cohen's $d_z = t/\sqrt{n}$ to characterise the practical magnitude of observed differences. For qualitative analysis, the three open-ended responses were reviewed independently by the research team, and recurring themes were identified through thematic grouping.

IV. Results and Analysis

A. Participant Demographics

As described in Section III-I, the final analysis sample comprised $n = 40$ valid responses after excluding two submissions.

Table II summarises the sample characteristics. The majority of participants were aged 21–25 (65.0%) and enrolled as university students (70.0%). Gender was approximately balanced, with 21 female (52.5%) and 19 male (47.5%)

TABLE II
PARTICIPANT DEMOGRAPHICS ($n = 40$).

Variable	Category	n	%
Age range	18–20	8	20.0
	21–25	26	65.0
	26–34	6	15.0
Gender	Female	21	52.5
	Male	19	47.5
Role	Student	28	70.0
	Working professional	12	30.0

respondents. Most participants reported low familiarity with health applications: 75.0% selected responses indicating rare or no use of such tools.

B. Quantitative Comparison

Each of the three COSMIN-grounded dimensions was measured by three Likert items (1–5 scale). Per-participant composite scores were computed by averaging the three items within each dimension. Table III presents per-item and composite means for both methods, together with paired t-test results ($df = 39$).

All three composite scores were significantly higher for the LLM-based method: Clarity ($\Delta = +0.13$, $p = .014$), Relevance ($\Delta = +0.42$, $p < .001$), and Quality ($\Delta = +0.29$, $p < .001$). Relevance showed the largest difference, while Clarity showed the smallest.

C. Clarity

Both methods were rated positively for Clarity, with all item means above 3.7. The LLM-based method scored slightly higher on each item: clear and easy to understand ($M = 4.22$ vs. 4.03), easy to know how to answer ($M = 4.10$ vs. 4.03), and specific and unambiguous wording ($M = 3.83$ vs. 3.70). However, these item-level differences were small, ranging from $+0.07$ to $+0.20$. The composite Clarity score was significantly higher for the LLM-based method ($t(39) = 2.58$, $p = .014$), but the small difference indicates that both versions were generally understandable. Wording specificity was the lowest-rated clarity item for both methods, suggesting a shared design challenge rather than an issue unique to either approach.

Open-ended responses supported this interpretation. Most respondents who answered the clarity question reported no confusion, with only isolated comments mentioning timeframe ambiguity, a need for clearer onboarding, or LLM-based questions feeling longer. This suggests that the rule-based method was not unclear; instead, the LLM-based method offered a modest clarity advantage while preserving overall comprehensibility.

D. Relevance

Relevance showed the clearest separation between methods. The LLM-based method was rated higher on all three relevance items: relevance to Sophie's health profile ($M =$ 4.15 vs. 3.77), addressing what seemed most important ($M =$ 4.00 vs. 3.62), and reflecting changes since the previous month ($M = 4.10$ vs. 3.60). The largest item-level gap was therefore the change-tracking item ($\Delta = +0.50$), which directly reflects the LLM method's ability to use Sophie's longitudinal context.

The composite Relevance score also showed the largest overall difference among the three dimensions ($\Delta = +0.42$, $t(39) = 3.69$, $p < .001$). This result aligns with the intended distinction between the two approaches. The rule-based method could ask clear predefined questions, but it could not automatically connect Sophie's Month 2 questions with her Month 1 profile unless those pathways were manually encoded. The LLM-based method was therefore perceived as more relevant because it could adapt question wording and selection to the persona's previous state.

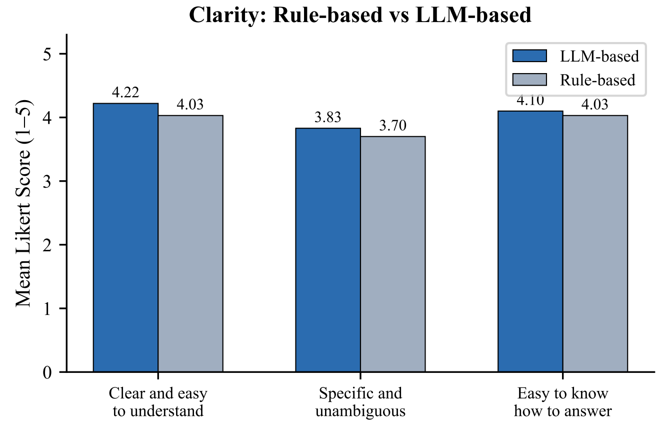

Fig. 1. Per-item Clarity scores for the LLM-based and rule-based methods ($n = 40$). All items were rated above 3.7, with small differences between methods.

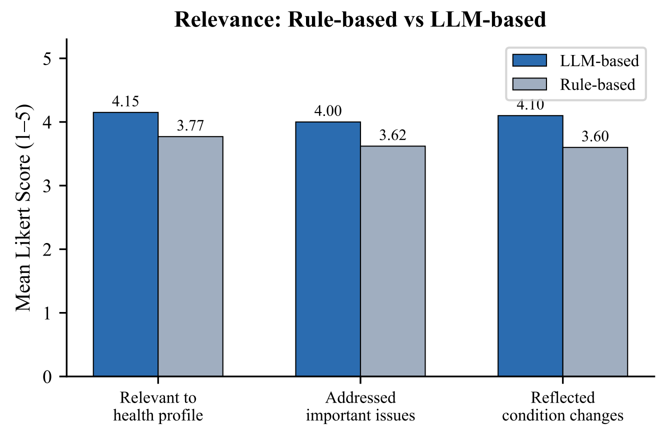

Fig. 2. Per-item Relevance scores ($n = 40$). The LLM-based method showed the largest advantage on the change-tracking item ($\Delta = +0.50$).

E. Quality

Quality ratings also favoured the LLM-based method, though the pattern was more specific than a general improvement in user experience. The LLM-based method scored

TABLE III
PER-ITEM AND COMPOSITE LIKERT SCORES (MEAN $\pm$ SD) WITH PAIRED t-TEST RESULTS ($n = 40$). EFFECT SIZE $d_z = t/\sqrt{n}$. ASTERISKS: $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$.

Dimension	Item	LLM (M $\pm$ SD)	Rule (M $\pm$ SD)	Δ	t	p	d_z
Clarity	Clear and easy to understand	4.22 ± 0.73	4.03 ± 0.95	+0.20	1.75	.088	0.28
	Specific and unambiguous	3.83 ± 0.96	3.70 ± 1.07	+0.12	1.30	.200	0.21
	Easy to know how to answer	4.10 ± 0.96	4.03 ± 1.05	+0.07	0.62	.538	0.10
	Composite	$\mathbf{4.05 \pm 0.73}$	$\mathbf{3.92 \pm 0.84}$	**+0.13**	**2.58**	**.014***	**0.41**
Relevance	Relevant to Sophie's health profile	4.15 ± 0.66	3.77 ± 0.80	+0.38	3.55	.001**	0.56
	Addressed important aspects	4.00 ± 0.75	3.62 ± 0.87	+0.38	2.94	.005**	0.46
	Reflected changes since prev. month	4.10 ± 0.78	3.60 ± 0.93	+0.50	3.29	.002**	0.52
	Composite	$\mathbf{4.08 \pm 0.63}$	$\mathbf{3.67 \pm 0.77}$	**+0.42**	**3.69**	**< .001*****	**0.58**
Quality	Captured important details	3.90 ± 0.74	3.50 ± 0.88	+0.40	3.57	.001**	0.56
	Well designed for meaningful assessment	4.00 ± 0.78	3.58 ± 0.93	+0.42	3.60	.001**	0.57
	Natural and appropriate	3.98 ± 0.89	3.92 ± 0.80	+0.05	0.36	.720	0.06
	Composite	$\mathbf{3.96 \pm 0.65}$	$\mathbf{3.67 \pm 0.73}$	**+0.29**	**3.76**	**< .001*****	**0.59**

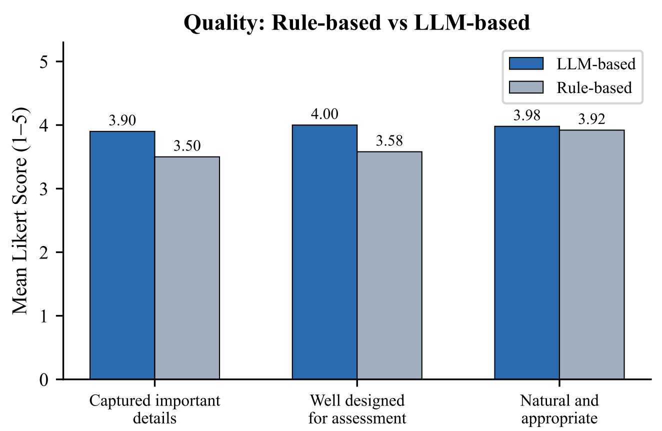

Fig. 3. Per-item Quality scores ($n = 40$). The natural-and-appropriate item was nearly tied ($\Delta = +0.06$), while other items showed larger differences.

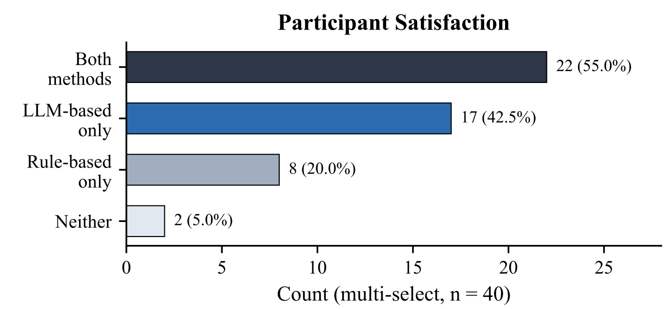

Fig. 4. Participant satisfaction endorsements (multi-select, $n = 40$). A majority endorsed both methods, with stronger endorsement for the LLM-based version.

higher for capturing important details ($M = 3.90$ vs. 3.50) and being well designed for a meaningful assessment ($M = 4.00$ vs. 3.58). In contrast, the natural and appropriate item was almost tied ($M = 3.98$ vs. 3.92), showing that participants did not necessarily experience the LLM version as more comfortable or natural to complete.

The composite Quality score was significantly higher for the LLM-based method ($\Delta = +0.29$, $t(39) = 3.76$, $p < .001$). This suggests that participants viewed the LLM-based transformations as better aimed at Sophie's situation, particularly in terms of capturing meaningful details, rather than simply as a more pleasant interaction.

F. Overall Preference

The overall satisfaction item was multi-select, so responses indicate endorsement rather than mutually exclusive preference. A majority of participants endorsed satisfaction with both methods ($n = 22$, 55.0%). The LLM-based method was endorsed by 17 participants (42.5%), while the rule-based method was endorsed by eight participants (20.0%). Two participants (5.0%) selected neither.

These results show broad acceptance of both methods, but stronger endorsement for the LLM-based version. Importantly, the large "satisfied with both" group indicates that the static rule-based method was not perceived as poor. Rather, the LLM-based method appeared to offer an additional benefit, especially where participants noticed greater personalisation and awareness of prior history. Figure 4 illustrates the distribution.

G. Qualitative Findings

The open-ended responses helped explain why the LLM-based method was rated more favourably, particularly in Relevance. Table IV summarises the thematic codes from the comparison question ($n = 39$ non-blank responses). The largest group expressed no strong preference between methods (28.2%), while personalisation and relevance was the most cited reason for preferring the LLM method (20.5%). Longitudinal awareness (10.3%) and natural conversational tone (7.7%) further supported the LLM preference. Conversely, four participants (10.3%) valued the rule-based method's simplicity and transparency, describing it as more straightforward. Rep-

TABLE IV
THEMATIC ANALYSIS OF QUESTIONNAIRE PREFERENCE (Q2, $n = 39$).

Theme	n	%	Example
No strong preference	11	28.2	"Both were easy to complete"
Personalisation / Relevance	8	20.5	"Felt more specific and tailored"
Longitudinal awareness	4	10.3	"Linked back to Month 1"
Simplicity (Rule-based)	4	10.3	"Rule-based was straightforward"
Natural / Conversational	3	7.7	"Feels more human-like"
Brevity / Efficiency	2	5.1	"Less questions, straight to point"

TABLE V
IMPROVEMENT SUGGESTIONS FROM OPEN-ENDED FEEDBACK (Q3, $n = 13$).

Theme	n	%	Example
More personalisation	4	30.8	"More depth in LLM questions"
Scoring / display bugs	2	15.4	"Scores seemed inverted"
New features (open text)	2	15.4	"Short answers for updates"
UI / navigation issues	1	7.7	"Side-by-side was hard to navigate"
Clearer instructions	1	7.7	"More guidance for the process"

resentative comments described the LLM version as "more specific and better tailored" and noted that it "linked back" to Month 1 information.

The qualitative responses also identified limits of the current prototype (Table V). Participants wanted deeper personalisation (7.7%), clearer instructions (5.1%), or more flexible input options such as open-text responses (2.6%). A small number of comments also noted scoring or display issues and navigation difficulty in the side-by-side comparison page. These comments indicate that participants valued the direction of LLM-based adaptation, but also expected a higher level of personal depth and interface polish from an adaptive system.

H. Summary of Key Observations

Across the nine evaluation items, the LLM-based questionnaire was rated equal to or higher than the rule-based questionnaire in every case. The largest differences appeared in Relevance and Quality, while the Clarity difference was smaller. This pattern suggests that both methods can produce understandable questions, but the LLM-based method was better at making those questions feel connected to Sophie's profile and previous-month condition. The qualitative themes converged with this interpretation: participants most often praised the LLM method for personalisation, history awareness, and contextual fit, while still recognising that the rule-based method was straightforward and usable.

V. DISCUSSION

A. Static vs Dynamic Adaptation

The central interpretation of this study is not that LLM-based questionnaires are categorically better than rule-based questionnaires. Rather, the meaningful contrast in Navigator is between a static branching design and a dynamically context-aware design. The rule-based method used predefined question text and fixed branching rules. This made it transparent and predictable, but also limited its ability to respond to Sophie's previous-month profile unless each possible pathway had been anticipated by the designer. By contrast, the LLM-based method used the same assessment context to rewrite, prioritise, or add questions that referred more directly to Sophie's profile and longitudinal state.

This distinction helps explain the observed pattern of results. The LLM-based method was perceived most strongly in the Relevance dimension, where participants evaluated whether questions reflected Sophie's persona, current situation, and previous-month condition. In comparison, the difference in Clarity was smaller, suggesting that carefully designed rule-based questions can still be clear and understandable. The main value of the LLM-based approach in this prototype is therefore not surface-level wording improvement, but the ability to scale context-aware adaptation without requiring every possible question pathway to be manually authored in advance. This interpretation is consistent with the broader motivation for adaptive assessment and LLM-supported patient-reported outcome measures [4], [8].

B. Interpretation of Research Questions

For RQ1, the findings provide only cautious support for the LLM-based method. Both versions were rated positively for Clarity, and the difference between methods was the smallest among the three evaluation dimensions. This suggests that clarity depends strongly on the quality of question writing, not only on whether a questionnaire is adapted with an LLM. A static rule-based method can remain clear when its items are carefully designed.

For RQ2, the findings suggest that participants perceived the LLM-based method as somewhat higher in Quality. This may reflect the way the LLM-adapted questions were framed around Sophie's student background and ongoing mental health pattern. However, this result should be interpreted as perceived questionnaire quality rather than evidence of clinical validity. The study did not evaluate whether either method produced more accurate mental health assessment outcomes.

For RQ3, the findings provide the strongest support for the LLM-based method. Relevance is the dimension most directly connected to contextual adaptation, and participants appeared to recognise when questions reflected Sophie's previous-month condition or current circumstances. This supports the argument that LLMs may be particularly useful when questionnaire systems need to connect standard assessment content with individual history, rather than simply present the same static items at each check-in.

Taken together, the improvements in perceived relevance and quality carry practical implications beyond user experience. If an adaptive method can produce questions that users find more relevant to their situation, respondents are more likely to engage meaningfully rather than providing superficial or habitual answers. In the context of health monitoring, more engaged and Thoughtful responses can improve the quality of collected data, which in turn can strengthen the accuracy of downstream risk prediction. While this study did not directly measure prediction outcomes, the observed pattern suggests that LLM-based adaptation may contribute to better data collection, and therefore, better prediction capability, by asking questions that users perceive as genuinely connected to their circumstances.

C. Limitations

Several limitations should be considered when interpreting these results. First, the study used a fixed presentation order: participants completed the rule-based version before the LLM-based version. This design ensured that all participants experienced the same comparison sequence, but it also introduces a possible order effect. Participants may have rated the second method more favourably because it appeared later, felt more novel, or was implicitly perceived as an improved version of the first method. Relatedly, the label "LLM-based" may have created an expectation that the AI version would be more advanced, which could have influenced subjective ratings independently of the actual question content.

Second, the prototype exposure was intentionally short. Each method presented only five monthly check-in questions, and the full study was designed to take approximately 10–15 minutes. This kept the task practical for anonymous online participation, but it may also have limited participants' ability to experience strong differences between the two methods. A longer questionnaire or repeated monthly use might reveal clearer differences in adaptation, but it would also increase participant burden. This is an important design trade-off: prior work on digital self-report has shown that repeated assessment can face completion and response-rate challenges [3], while computerised adaptive testing research has specifically aimed to reduce item burden in mental health assessment [5], [6].

Third, the study used a fixed persona rather than participants' own health data. The Sophie scenario improved experimental control because every participant evaluated the same profile, but it reduced ecological validity. Participants were judging whether questions felt appropriate for a fictional case, not whether they felt personally understood by the system. Real longitudinal use may produce different reactions, especially when questions refer to sensitive personal experiences.

Fourth, the study measured perceived clarity, quality, relevance, and preference rather than clinical accuracy or health outcomes. The results therefore cannot show that the LLM-based method leads to improved clinical outcomes or more accurate risk assessment. Moreover, the prototype did not log the risk assessment scores (derived from PHQ-9 and GAD-7 mappings) computed during each condition, preventing a direct comparison of whether the two methods produced different risk profiles for the same persona scenario. Future work should record these scores to enable analysis of whether improved question relevance translates into measurably different assessment outcomes.

Fifth, the prototype focused only on the mental health pathway, used a single LLM provider, and was evaluated with a small convenience sample. Future studies should use counterbalanced presentation orders, larger and more diverse participant groups, longer longitudinal scenarios, and clinician review of LLM-adapted questions before making claims about clinical suitability.

VI. Conclusion

This study evaluated whether an LLM-based adaptive questionnaire method could improve user perceptions of mental health check-in questions compared with a static rule-based method. Using the Navigator prototype for mental health and a controlled Sophie persona, participants completed both versions and rated them across Clarity, Relevance, and Quality. The findings show that the LLM-based method was rated higher across all three composite dimensions, with the clearest advantage in Relevance. Qualitative responses supported this pattern by highlighting personalisation, history awareness, and contextual fit as the main reasons participants preferred the LLM-based version.

The study does not show that LLMs are clinically superior to rule-based questionnaires, nor that they produce more accurate mental health risk assessment. Overall, this study demonstrates that LLM-based adaptation can make questionnaire content feel more connected to a user's profile and previous responses within a controlled prototype setting. The main contribution of this work is therefore a human-centred evaluation of static versus dynamic adaptation, showing that the value of LLMs in mental health questionnaire systems lies primarily in contextual relevance rather than clearer wording alone.

A. Future Work

Future work should extend this evaluation in several ways. First, a counterbalanced study design should be used so that some participants complete the LLM-based method first and others complete the rule-based method first, allowing order effects to be separated from method effects. Second, the study should be repeated with larger and more diverse samples, including older participants and people with more varied experience using health applications. Third, the prototype should be evaluated longitudinally across multiple monthly check-ins using real participant data, subject to appropriate ethics approval, to test whether perceived relevance remains meaningful over time.

Future versions should also record downstream risk scores and compare whether different question sequences produce measurably different assessment outcomes. This would connect the present user-perception findings to the broader Navigator aim of improving data quality for health monitoring and prediction. Making individual check-in questions optional and tracking skip rates per method would further provide a behavioural measure of perceived relevance, complementing the self-reported ratings used in this study: a question that participants consistently skip may indicate low relevance more directly than a Likert rating alone. Finally, LLM-based or LLM-adapted questions should be reviewed by clinical or domain experts before use in higher-risk settings, and the approach should be tested across other health domains beyond mental health.

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